

multidimensional gradient descent

CSCI
270



if at first you don't succeed



try try again

you are **zero** years old



i will guess your age
or you win a **prize**



ooo

$$y = \theta x$$
$$\theta = 3.5$$



rather than estimating age
based on cs experience



you will do it based on hair

as people age,
their hair color
tends to get
lighter and less
saturated

		(Blue)				
L	S	1	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	0
1						
$\frac{7}{8}$						
$\frac{3}{4}$						
$\frac{5}{8}$						
$\frac{1}{2}$						
$\frac{3}{8}$						
$\frac{1}{4}$						
$\frac{1}{8}$						
0						

↑
lightness

← saturation

again you train a model based on
data collected from your friends



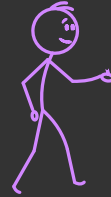
alice
age 20

saturation: .72
lightness: .06



bob
age 28

saturation: .34
lightness: .25



chandra
age 41

saturation: .17
lightness: .57

again you train a model based on
data collected from your friends

	x_s saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

model space



loss function



optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

training data

	x_S saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

test data



model space

$$y = \theta_s x_s + \theta_L x_L$$

loss function



optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

training data

	x_s saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

test data



model space

$$y = \theta_s x_s + \theta_L x_L$$

loss function

$$\text{want: } \begin{cases} 20 \approx .72 \theta_s + .06 \theta_L \\ 28 \approx .34 \theta_s + .25 \theta_L \\ 41 \approx .17 \theta_s + .57 \theta_L \end{cases}$$

training data

	x_s saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

test data



model space

$$y = \theta_s x_s + \theta_L x_L$$

loss function

$$\text{want: } \begin{cases} (.72\theta_s + .06\theta_L - 20)^2 \approx 0 \\ (.34\theta_s + .25\theta_L - 28)^2 \approx 0 \\ (.17\theta_s + .57\theta_L - 41)^2 \approx 0 \end{cases}$$

training data

	x_s saturation	x_L lightness	y age
	.72	.06	0
	.34	.25	28
	.17	.57	41

optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

test data



model space

$$y = \theta_s x_s + \theta_L x_L$$

training data

	x_s saturation	x_L lightness	y age
	.72	.06	0
	.34	.25	28
	.17	.57	41

loss function

want:

$$\begin{aligned} & (.72\theta_s + .06\theta_L - 20)^2 \\ & + (.34\theta_s + .25\theta_L - 28)^2 \\ & + (.17\theta_s + .57\theta_L - 41)^2 \end{aligned}$$

close to zero

optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

test data



model space

$$y = \theta_s x_s + \theta_L x_L$$

training data

	x_s saturation	x_L lightness	y age
	.72	.06	0
	.34	.25	28
	.17	.57	41

loss function

$$L(\theta_s, \theta_L) = \begin{aligned} & (.72\theta_s + .06\theta_L - 20)^2 \\ & + (.34\theta_s + .25\theta_L - 28)^2 \\ & + (.17\theta_s + .57\theta_L - 41)^2 \end{aligned}$$

optimization

$$\underset{\theta}{\operatorname{argmin}} L(\theta)$$

test data



model space

$$y = \theta_s x_s + \theta_L x_L$$

training data

	x_s saturation	x_L lightness	y age
	.72	.06	0
	.34	.25	28
	.17	.57	41

loss function

$$L(\theta_s, \theta_L) = \begin{aligned} & (.72\theta_s + .06\theta_L - 20)^2 \\ & + (.34\theta_s + .25\theta_L - 28)^2 \\ & + (.17\theta_s + .57\theta_L - 41)^2 \end{aligned}$$

optimization

$$\underset{\theta_s, \theta_L}{\operatorname{argmin}} L(\theta_s, \theta_L)$$

test data



but how do we
minimize a
multivariable
function?

loss function

$$L(\theta_s, \theta_L) = \begin{aligned} & (.72\theta_s + .06\theta_L - 20)^2 \\ & + (.34\theta_s + .25\theta_L - 28)^2 \\ & + (.17\theta_s + .57\theta_L - 41)^2 \end{aligned}$$

optimization

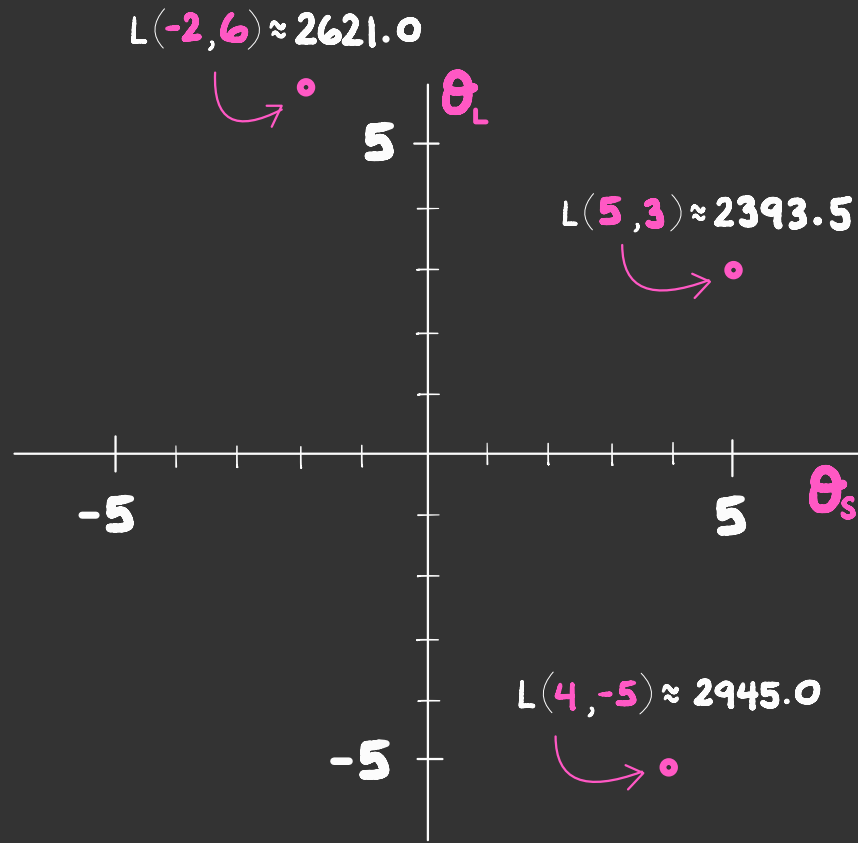
$$\underset{\theta_s, \theta_L}{\operatorname{argmin}} L(\theta_s, \theta_L)$$



we want to find the
point that minimizes
the loss

loss function

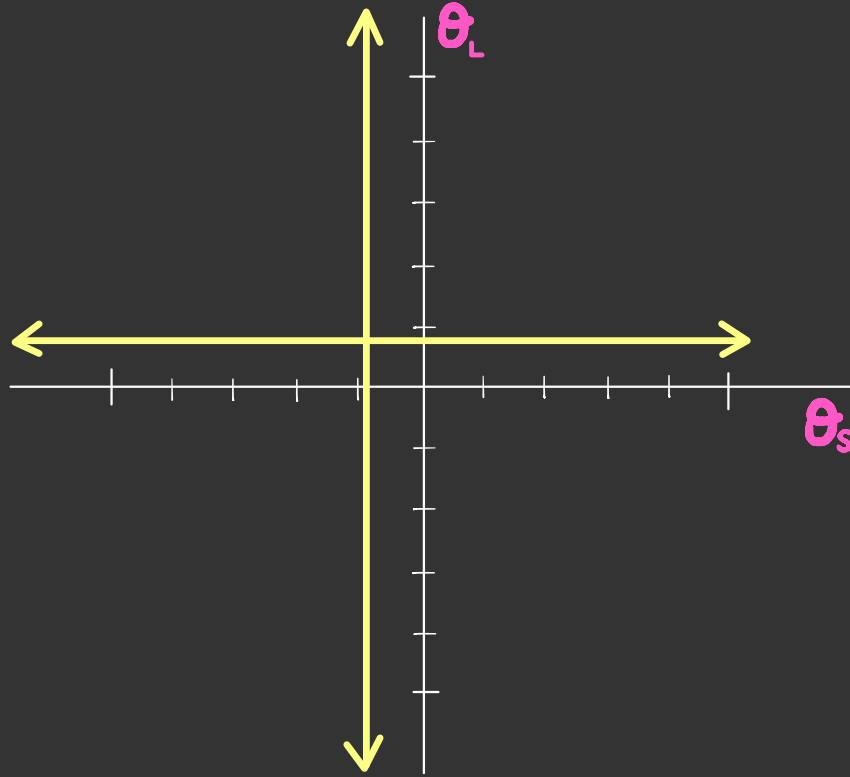
$$L(\theta_s, \theta_L) = \begin{aligned} & (.72\theta_s + .06\theta_L - 20)^2 \\ & + (.34\theta_s + .25\theta_L - 28)^2 \\ & + (.17\theta_s + .57\theta_L - 41)^2 \end{aligned}$$



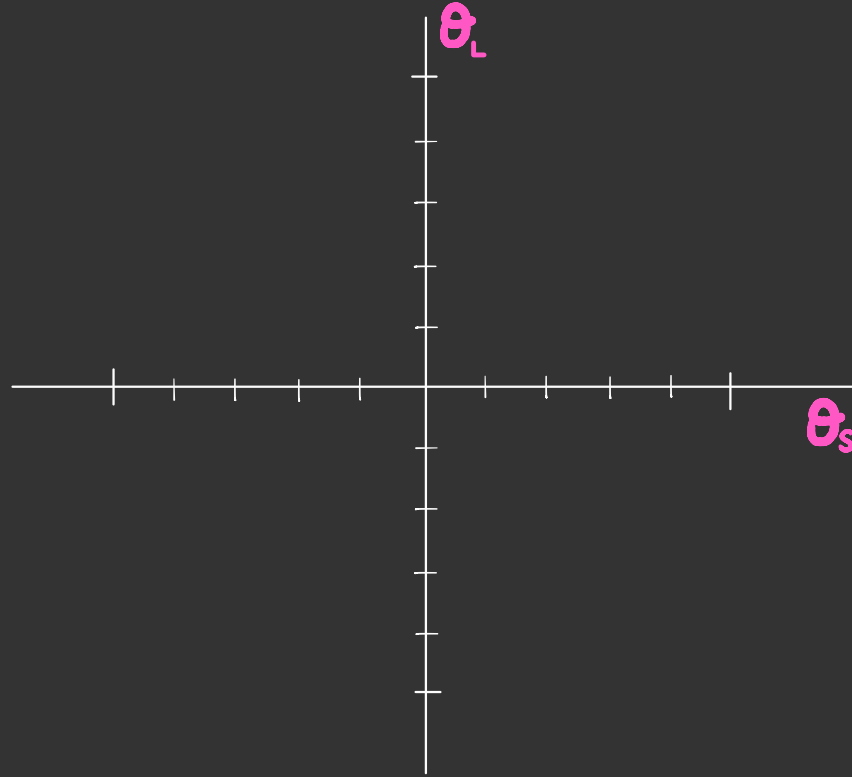
but how do
we minimize
two variables
simultaneously?

loss function

$$L(\theta_s, \theta_L) = \begin{aligned} & (.72\theta_s + .06\theta_L - 20)^2 \\ & + (.34\theta_s + .25\theta_L - 28)^2 \\ & + (.17\theta_s + .57\theta_L - 41)^2 \end{aligned}$$

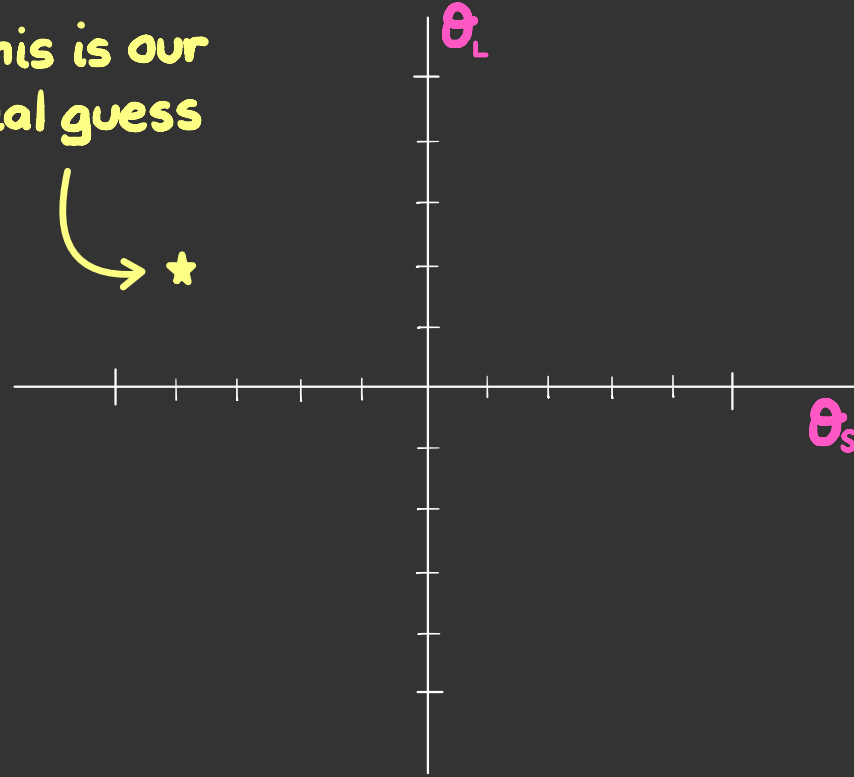


pretend the other
variable is already
at its ideal value



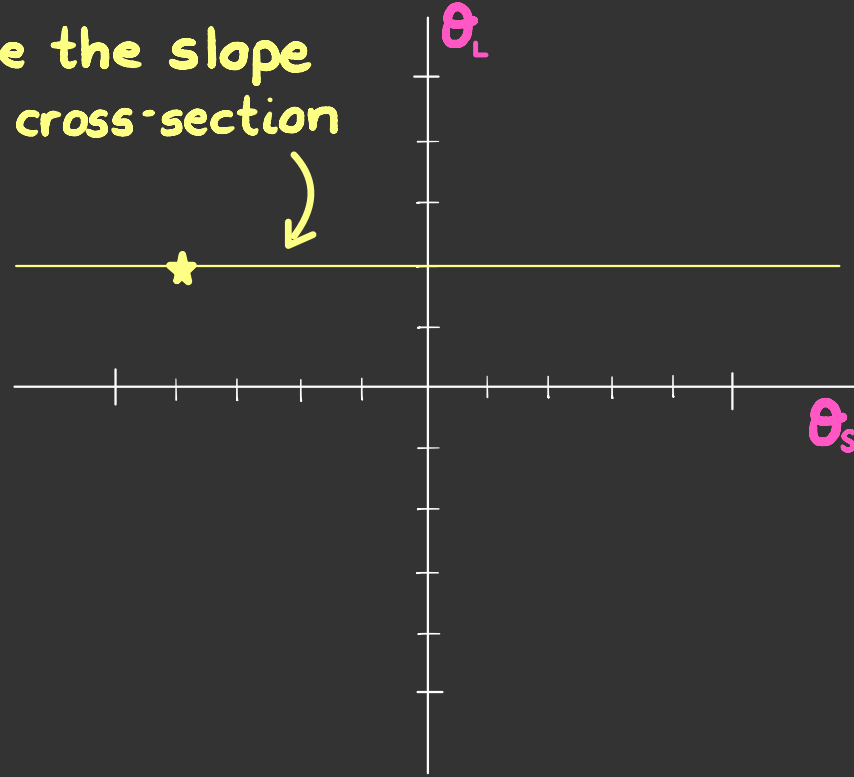
pretend the other
variable is already
at its ideal value

if this is our
initial guess



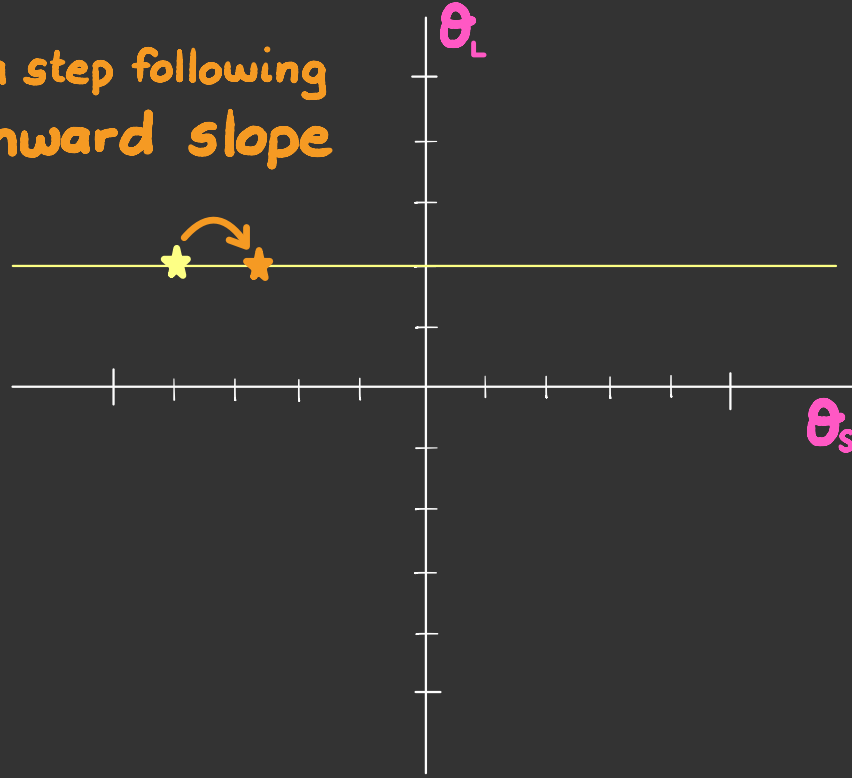
pretend the other
variable is already
at its ideal value

determine the slope
along this cross-section



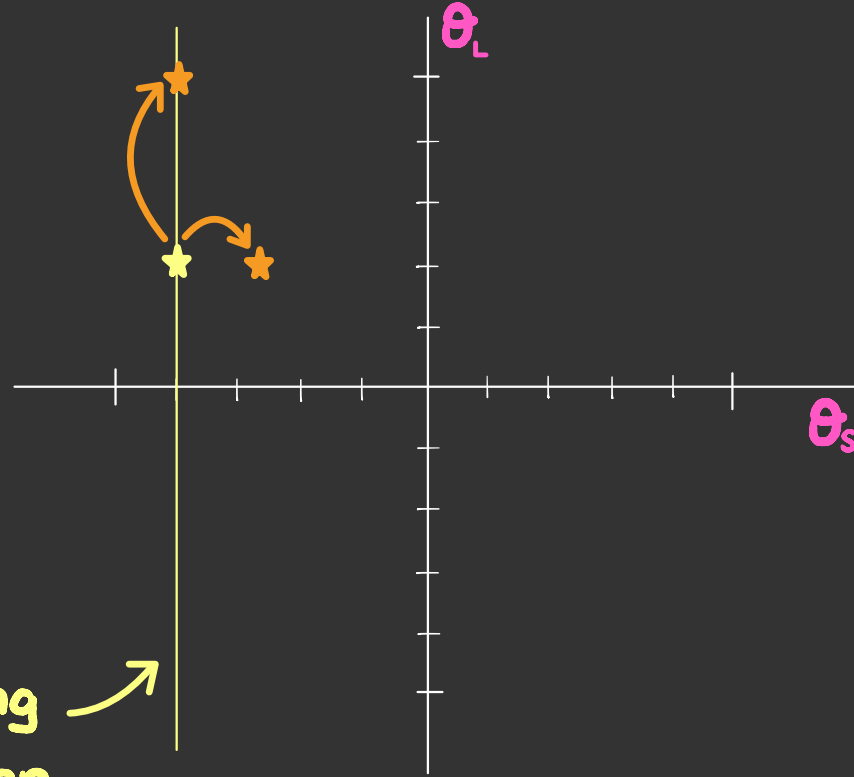
and take a step following
the downward slope

pretend the other
variable is already
at its ideal value

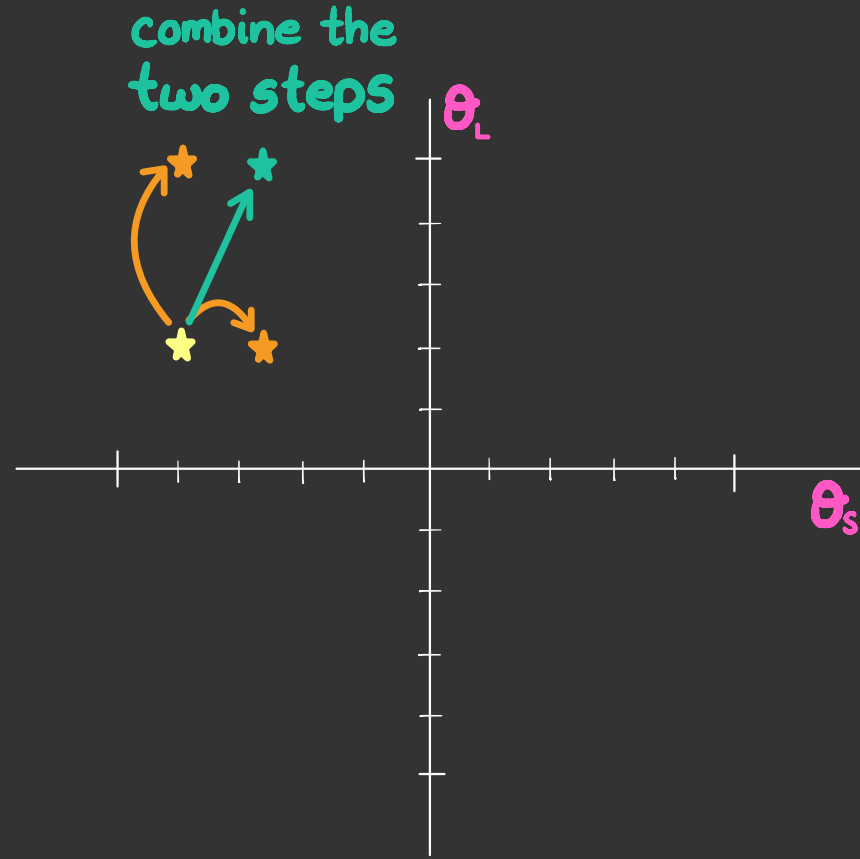


pretend the other
variable is already
at its ideal value

do the same along
this cross-section

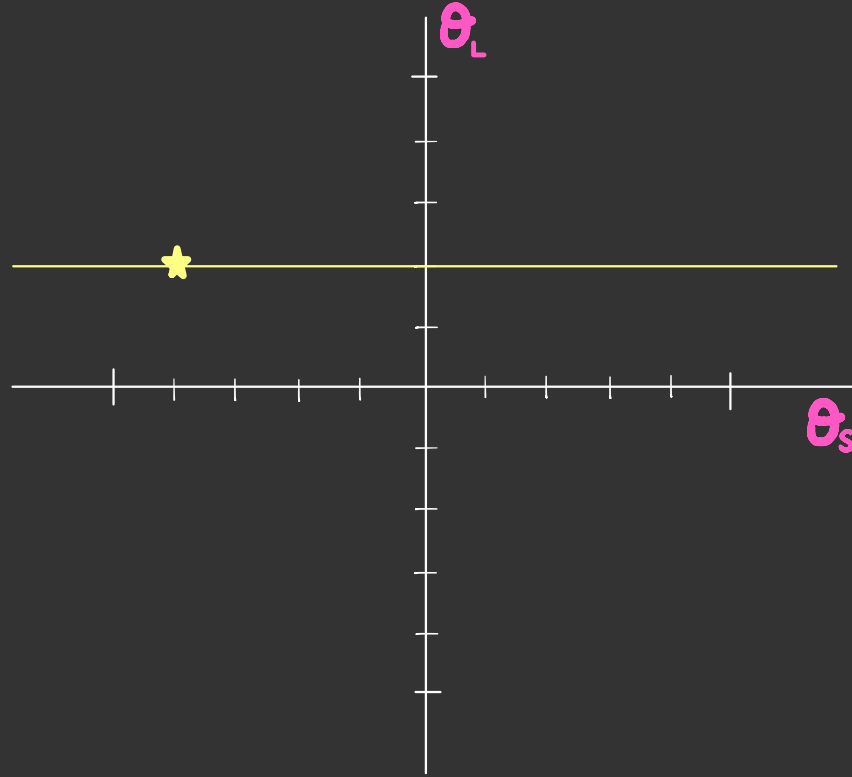


pretend the other
variable is already
at its ideal value



"partial derivative"

pretend the other
variables are already
at their ideal values
and compute the
slope along the
cross-section



"partial derivative"

pretend the other
variables are already
at their ideal values
and compute the
slope along the
cross-section

$$\begin{aligned}\frac{\partial L}{\partial \theta_s} &= \frac{\partial L}{\partial \theta_s} ((.72\theta_s + .06\theta_L - 20)^2) \\ &\quad + \frac{\partial L}{\partial \theta_s} ((.34\theta_s + .25\theta_L - 28)^2) \\ &\quad + \frac{\partial L}{\partial \theta_s} ((.17\theta_s + .57\theta_L - 41)^2) \\ &= 2(.72\theta_s + .06\theta_L - 20) \frac{\partial L}{\partial \theta_s} (.72\theta_s + .06\theta_L - 20) \\ &\quad + 2(.34\theta_s + .25\theta_L - 28) \frac{\partial L}{\partial \theta_s} (.34\theta_s + .25\theta_L - 28) \\ &\quad + 2(.17\theta_s + .57\theta_L - 41) \frac{\partial L}{\partial \theta_s} (.17\theta_s + .57\theta_L - 41) \\ &= 2(.72\theta_s + .06\theta_L - 20)(.72) \\ &\quad + 2(.34\theta_s + .25\theta_L - 28)(.34) \\ &\quad + 2(.17\theta_s + .57\theta_L - 41)(.17) \\ &= 1.3258\theta_s + .4502\theta_L - 61.78\end{aligned}$$

"partial derivative"

pretend the other
variables are already
at their ideal values
and compute the
slope along the
cross-section

$$\begin{aligned}\frac{\partial L}{\partial \theta_L} &= \frac{\partial L}{\partial \theta_L} ((.72\theta_S + .06\theta_L - 20)^2) \\ &+ \frac{\partial L}{\partial \theta_L} ((.34\theta_S + .25\theta_L - 28)^2) \\ &+ \frac{\partial L}{\partial \theta_L} ((.17\theta_S + .57\theta_L - 41)^2) \\ &= 2(.72\theta_S + .06\theta_L - 20) \frac{\partial L}{\partial \theta_L} (.72\theta_S + .06\theta_L - 20) \\ &+ 2(.34\theta_S + .25\theta_L - 28) \frac{\partial L}{\partial \theta_L} (.34\theta_S + .25\theta_L - 28) \\ &+ 2(.17\theta_S + .57\theta_L - 41) \frac{\partial L}{\partial \theta_L} (.17\theta_S + .57\theta_L - 41) \\ &= 2(.72\theta_S + .06\theta_L - 20)(.06) \\ &+ 2(.34\theta_S + .25\theta_L - 28)(.25) \\ &+ 2(.17\theta_S + .57\theta_L - 41)(.57) \\ &= .4502\theta_S + .782\theta_L - 63.14\end{aligned}$$

it is often convenient to collect the partial derivatives into a **gradient vector**

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial \theta_s} \\ \frac{\partial L}{\partial \theta_L} \end{bmatrix} = \begin{bmatrix} 1.3258 \theta_s + .4502 \theta_L - 61.78 \\ .4502 \theta_s + .782 \theta_L - 63.14 \end{bmatrix}$$

the pseudocode for gradient descent
generalizes easily to multiple variables

gradient descent

(single variable)

► choose initial position $\theta^{(0)}$

► for $t \in \{0, \dots, T-1\}$:

$$\theta^{(t+1)} = \theta^{(t)} - \alpha^{(t)} \frac{dL}{d\theta}(\theta^{(t)})$$

► return $\theta^{(T)}$

the pseudocode for gradient descent
generalizes easily to multiple variables

gradient descent

(multivariable)

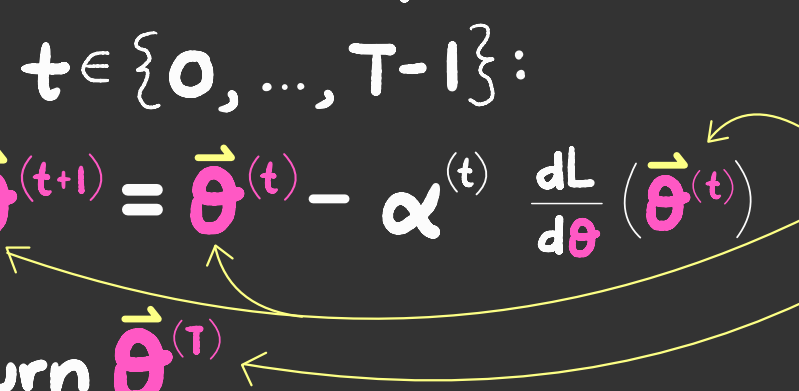
► choose initial position $\vec{\theta}^{(0)}$

► for $t \in \{0, \dots, T-1\}$:

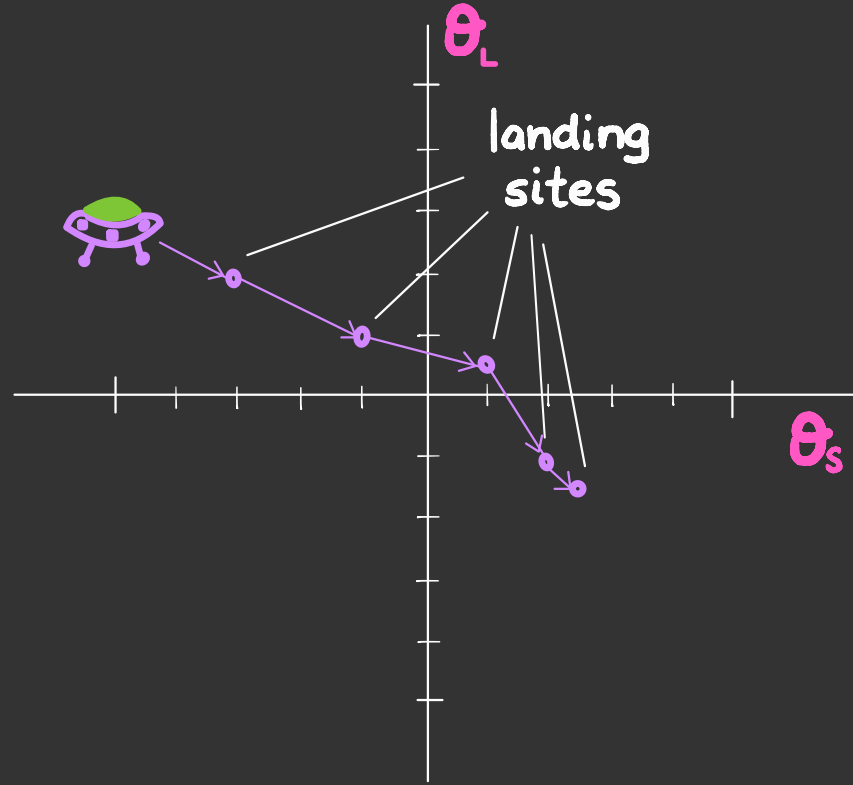
$$\vec{\theta}^{(t+1)} = \vec{\theta}^{(t)} - \alpha^{(t)} \frac{dL}{d\theta}(\vec{\theta}^{(t)})$$

► return $\vec{\theta}^{(T)}$

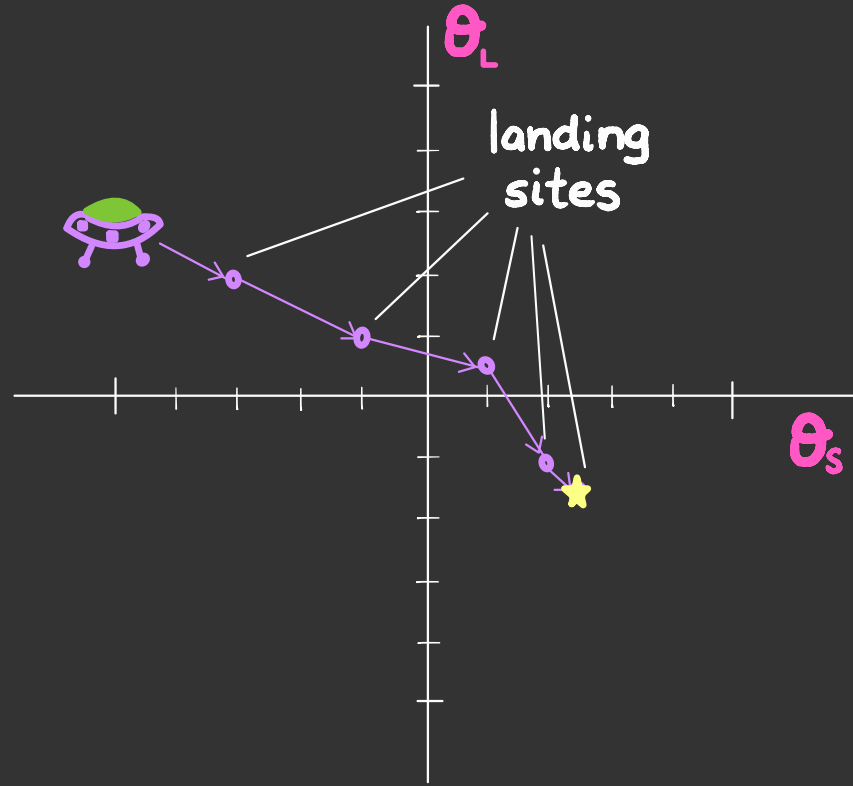
these
are
now
vectors



birds-eye view
of a 2-dimensional
gradient descent

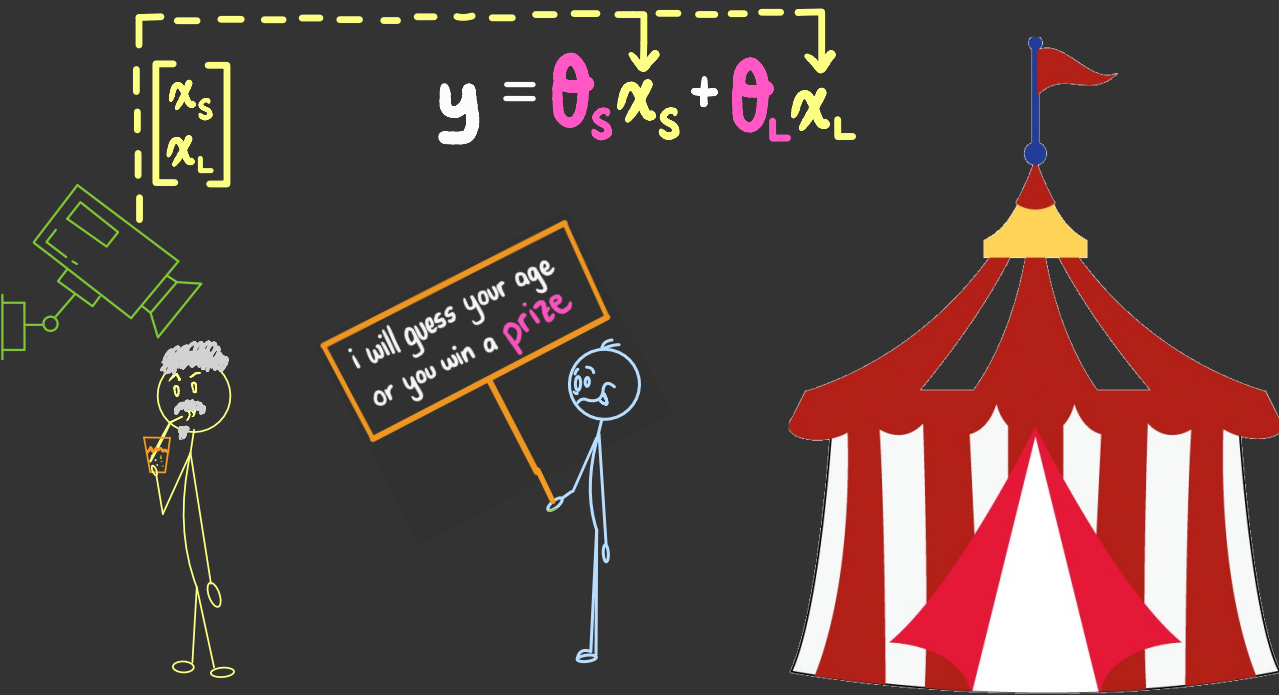


once you find values
of θ_s and θ_L
with acceptably
small loss





you set up a hidden camera that measures
the saturation and lightness of hair



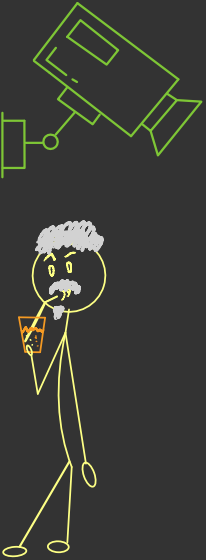
then you predict the customer's age with your trained model

you are
57

i will guess your age
or you win a prize



success!





! will guess your age
or you win a **prize** !

